

Social Media & User Behaviour Analysis Report

SQL + Python Analysis |

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Platform Statistics Overview

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Social Commerce & Churn Risk

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SQL Analysis — Platform Queries (Q1–Q15)

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Agenda

Q1 — Total Platforms Count

```
select
| COUNT(platform) as total_platform
from
| platform_statistics_2026;
```

Result: 10 platforms

10 major social media platforms tracked in 2026 database.

Q2 — Highest Monthly Active Users

```
SELECT
  top 1 platform,
  monthly_active_users_billions
from
  platform_statistics_2026
order by
  monthly_active_users_billions DESC;
```

Result: Facebook — 3.07B users

Facebook remains #1 with 3.07 billion MAU globally.

Q3 — Avg Daily Time Across Platforms

```
SELECT
| AVG(avg_daily_time_minutes) as average_daily_time
FROM
| platform_statistics_2026;
```

Result: ~52 minutes/day

Users spend on average 52 minutes daily across all platforms.

Platform Growth & Engagement — Q4, Q5, Q6

Q4 — Fast Growing Platforms (YoY > 10%)

```
select
| platform,
| year_over_year_growth_pct
from
| platform_statistics_2026
where
| year_over_year_growth_pct > 10;
```

Result: TikTok (18.2%), Instagram (12.5%), Threads (11.1%)

Short-video & niche platforms driving highest growth YoY.

Q5 — Platforms by Primary Age Group

```
SELECT
| primary_age_group,
| COUNT(platform) as total_platform
from
| platform_statistics_2026
group by
| primary_age_group
order by
| total_platform desc;
```

Result: 18–34 (5 platforms), 25–44 (3), 35–54 (2)

Most platforms target the 18–34 age bracket.

Q6 — Top 3 by Engagement Rate

```
SELECT
| top 3 platform,
| avg_engagement_rate_pct
from
| platform_statistics_2026
order by
| avg_engagement_rate_pct desc;
```

Result: TikTok (8.5%), Instagram (4.7%), YouTube (3.8%)

TikTok leads engagement by wide margin due to video algorithm.

Q7 — Total Users in Dataset

```
SELECT
| count(*) as total_users
from
| social_media_user_behavior;
```

Result: 5,000 users

Dataset comprises 5,000 diverse social media users globally.

Q8 — Avg Daily Screen Time

```
select
| AVG(daily_screen_time_minutes) as average_time_spent
from
| social_media_user_behavior;
```

Result: ~147 minutes/day

Users average ~2.5 hours/day on social media — well above platform averages.

Q9 — Top 5 Users by Followers

```
select
| top 5 user_id,
| followers_count
from
| social_media_user_behavior
order by
| followers_count desc;
```

Result: Top user ~980K followers

Power users drive disproportionate reach — top 5 collectively hold millions of followers.

Q10 — Users by Country

```
SELECT
| country,
| COUNT(user_id) as total_users
from
| social_media_user_behavior
group by
| country
order by
| total_users desc;
```

Top: USA (812), India (743), UK (521)

USA & India dominate user base — key markets for social commerce.

Q11 — High-Value Social Buyers

```
select
| *
from
| social_media_user_behavior
where
| has_purchased_via_social = 1
| AND monthly_social_spending_usd > 0.7;
```

Result: ~2,100 qualified buyers

42% of users both bought via social and have measurable spend.

Q12 — Churn Risk Users

```
select
| COUNT(*) as churned_users
FROM
| (
|   select
|   | case
|   |   when daily_screen_time_minutes < 30
|   |   and monthly_social_spending_usd = 0 then 1
|   |   else 0
|   | end as churn_risk
|   from
|   | social_media_user_behavior
| ) t
where
| churn_risk = 1;
```

Result: ~680 high-churn-risk users

13.6% of users show churn signals — low time + zero spend.

Q13 — Avg Satisfaction by Platform

```
SELECT
  primary_platform,
  AVG(
    case
      when platform_satisfaction = 'very satisfied' then 5
      when platform_satisfaction = 'satisfies' then 4
      when platform_satisfaction = 'neutral' then 3
      when platform_satisfaction = 'dissatisfies' then 2
      when platform_satisfaction = 'very dissatisfied' then 1
    end
  ) as avg_satisfaction_score
from
  social_media_user_behavior
group by
  primary_platform
order by
  avg_satisfaction_score desc;
```

Top: YouTube (4.2), Instagram (3.9), TikTok (3.8)

YouTube users most satisfied; Twitter/X shows lowest satisfaction scores.

Q14 — User vs Platform Engagement Gap

```
SELECT
  u.primary_platform,
  AVG(u.engagement_rate_pct) as user_avg_engagement,
  p.avg_engagement_rate_pct as platform_engagement,
  AVG(u.engagement_rate_pct) - p.avg_engagement_rate_pct as engagement_gap
from
  social_media_user_behavior u
  join platform_statistics_2026 p on u.primary_platform = p.platform
group by
  u.primary_platform,
  p.avg_engagement_rate_pct
order by
  engagement_gap desc;
```

LinkedIn gap: +1.8% | TikTok gap: -0.9%

LinkedIn users outperform platform avg; TikTok users underperform platform benchmark.

Q15 — Platforms Where User Time > Platform Avg

```
SELECT
  u.primary_platform AS platform,
  AVG(CAST(u.daily_screen_time_minutes AS FLOAT)) AS user_avg_time,
  p.avg_daily_time_minutes AS platform_avg_time
FROM
  social_media_user_behavior u
  JOIN platform_statistics_2026 p ON u.primary_platform = p.platform
GROUP BY
  u.primary_platform,
  p.avg_daily_time_minutes
HAVING
  AVG(CAST(u.daily_screen_time_minutes AS FLOAT)) > p.avg_daily_time_minutes;
```

Result: Instagram, TikTok, YouTube exceed platform avg

Users exceed platform-reported averages on 3 top platforms — engagement deeper than reported.

Gender Distribution

```
▶ # User demographics
print(df['gender'].value_counts())
print(df['age_group'].value_counts())
print(df['country'].value_counts())
```

Avg Daily Screen Time by Age Group (mins)

```
▶ # screen time analysis

print(df["daily_screen_time_minutes"].describe())
print(df.groupby("age_group")["daily_screen_time_minutes"].mean().sort_values(ascending=False))
```

Key Insight: Males slightly outnumber females (49.1% vs 46.2%). Younger users (18–24) spend 2× more time than 55+ cohort — targeted youth campaigns yield highest exposure.

Primary Platform Usage

```
▶ # platform popularity  
print(df["primary_platform"].value_counts())
```

Preferred Content Type

```
▶ # content preference  
print(df["preferred_content_type"].value_counts())  
print(df.groupby("primary_device")["preferred_content_type"].value_counts())
```

Short-form video dominates preferences (33.6%). Instagram & TikTok together hold 47% of primary platform share — ideal for video-first campaigns.

Engagement by Platform & Social Commerce

Avg Engagement Rate % by Platform

```
# Followers distribution
print(df[["user_id","followers_count"]].sort_values("followers_count", ascending=False).head(10))
print(df["followers_count"].describe())
```

```
# Engagement analysis

print(df.groupby("primary_platform")["engagement_rate_pct"].mean().sort_values(ascending=False))
print(df.groupby("is_content_creator")["engagement_rate_pct"].mean())
```

Monthly Social Spending by Income Bracket (USD)

```
# social commerce

print(df["has_purchased_via_social"].value_counts(normalize=True) * 100)
print(df.groupby("income_bracket")["monthly_social_spending_usd"].mean().sort_values(ascending=False))
```

Social Purchase Rate: 62.4% of users have purchased via social |
Non-buyers: 37.6%

High income users spend 7.6× more via social than low income.
TikTok Shop & Instagram Shopping are highest ROI channels.

Churn Risk Users by Age Group

```
▶ # churn risk analysis
df["churn_risk"] = ((df["daily_screen_time_minutes"] < 30) &
                    (df["monthly_social_spending_usd"] == 0))
print("Churn risk users:", df["churn_risk"].sum())
print(df[df["churn_risk"] == True]["age_group"].value_counts())
```

Influencer Followers vs Non-Followers

```
▶ # influence impact
print(df.groupby("follows_influencers")[["monthly_social_spending_usd", "engagement_rate_pct"]].mean())
print(df.groupby("follows_influencers")["platform_satisfaction"].value_counts(normalize=True))
```

Churn skews heavily toward older users (55+ = 35%). Influencer followers spend 2× more & engage 2× more — **influencer partnerships are high-ROI acquisition channels.**

Self-Reported Mental Health Effect

```
# cyber bullying
print(df["experienced_cyberbullying"].value_counts())
print(df.groupby("gender")["experienced_cyberbullying"].mean() * 100)
print(df.groupby("age_group")["experienced_cyberbullying"].mean() * 100)
```

Sleep Hours vs Addiction Level

```
# mental health impact
print(df["self_reported_mental_health_effect"].value_counts())
print(df.groupby("addiction_level_1_to_10")["sleep_hours_per_night"].mean())
```

Cyberbullying: 28.4% experienced it. Female users affected more (32.1%) than males (25.6%). Peaks in 18–24 age group (38.7%).

Higher addiction levels directly correlate with less sleep. Users at level 10 average 5.4 hrs vs 7.8 hrs at level 1 — a 30% reduction.

Business Insights & Recommendations

01 Double Down on Short-Form Video

TikTok (8.5% ER) & Instagram Reels outperform all other formats. Allocate 40%+ of content budget to short videos.

02 Target Youth 18–34 for Max Reach

53% of social audience is 18–22. Youth-first campaigns on TikTok/Instagram deliver highest engagement ROI.

03 Activate Social Commerce Funnel

62.4% have purchased via social. Integrate in-app checkout on Instagram & TikTok to reduce drop-off.

04 Re-engage Churn Risk Segment

680 high-risk users show low screen time + zero spend. Loyalty rewards & push notification campaigns can recover them.

05 Leverage Influencer Partnerships

Influencer followers spend 2× more. Partner with micro-influencers (10K–100K) for authentic, cost-effective reach.

06 Prioritize Mental Health Messaging

37% report negative mental health effects. Brands that promote digital wellness differentiate positively with Gen Z.



Thank You

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